

Max Kurzner

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Professional Summary

PhD candidate in astrophysics with deep experience building reliable data pipelines, curated datasets, derived metrics, and reproducible analyses from complex, heterogeneous source data. Strong Python, SQL, pandas, NumPy, scikit-learn, statistical modelling, and technical documentation background, with additional industry experience in healthcare analytics, corporate research, and actuarial modelling. Experienced in defining consistent analytical samples and metrics, validating outputs against source data, investigating discrepancies across transformation layers, and communicating data logic to technical and non-technical collaborators. Interested in analytics roles focused on trusted data foundations, metric consistency, self-serve reporting, and AI-enabled decision systems.

Core Skills

Analytics Engineering: SQL, data modelling, curated datasets, metric definitions, data validation, discrepancy investigation, documentation, reproducible transformations

Programming & Data: Python, pandas, NumPy, SciPy, scikit-learn, matplotlib, Astropy, Bash/Unix, Git, Jupyter

Statistical Analysis: Uncertainty propagation, Bayesian inference, Monte Carlo methods, model diagnostics, robustness checks

Machine Learning & AI Evaluation : PyTorch, TensorFlow/Keras, supervised learning, CNN transfer learning, subgroup evaluation, failure-mode analysis, AI-assisted workflow evaluation

Communication & Documentation: Stakeholder-facing analysis, technical writing, clear documentation of assumptions, data lineage, metric definitions, and AI-legible analytical context

Professional and Research Experience

University of Victoria — Graduate Researcher

Victoria, BC | September 2021 – present

- Built Python data pipelines to clean, join, validate, and analyze thousands of records from multiple source systems, including image data, human labels, measurement tables, and external catalogues.
- Created trusted datasets and derived metrics used for recurring analysis, reporting, publication figures, and collaborator decision-making.
- Investigated data discrepancies end-to-end by tracing issues from final reports back through source files, transformation logic, filtering rules, and intermediate datasets.

- Defined and documented metric logic, sample definitions, quality flags, assumptions, and caveats to make analyses consistent, auditable, and reusable.
- Built statistical workflows combining model outputs, human annotations, and structured data, with emphasis on uncertainty, robustness checks, and clear communication.
- Developed and evaluated machine learning workflows for image classification, including performance testing under noisy labels, limited supervision, domain shift, and failure modes.

Remedy Partners - *Clinical Analyst*

Norwalk, CT | October 2018 - August 2019

- Evaluated and developed analytical tools used by healthcare organizations to assess financial performance, operational outcomes, and risk in a data-driven healthcare environment.
- Translated ambiguous clinical and operational requirements into clear quantitative analyses for internal stakeholders, explicitly surfacing uncertainty, assumptions, and data-quality limitations.

Third Bridge Group - *Analyst*

New York, NY | May 2018 - September 2018

- Conducted primary research and competitive analysis to support client decisions on corporate strategy, product launches, market entry, and industry positioning.
- Rapidly synthesized input from subject-matter experts into concise, decision-relevant analyses under tight deadlines.

Tokio Millenium Re AG - *Actuarial Intern*

Stamford, CT | July 2017- November 2017

- Provided analytical support for probabilistic pricing and reserving models used to forecast loss development across insurance portfolios.
- Assessed model sensitivity to assumptions and historical data, supporting risk-aware decision-making under uncertainty.

Education

University of Victoria — PhD in Astrophysics, expected July 2026

Tufts University — Master of Science in Astrophysics, 2021

Colgate University — Bachelor of Arts in Astronomy & Physics, 2017